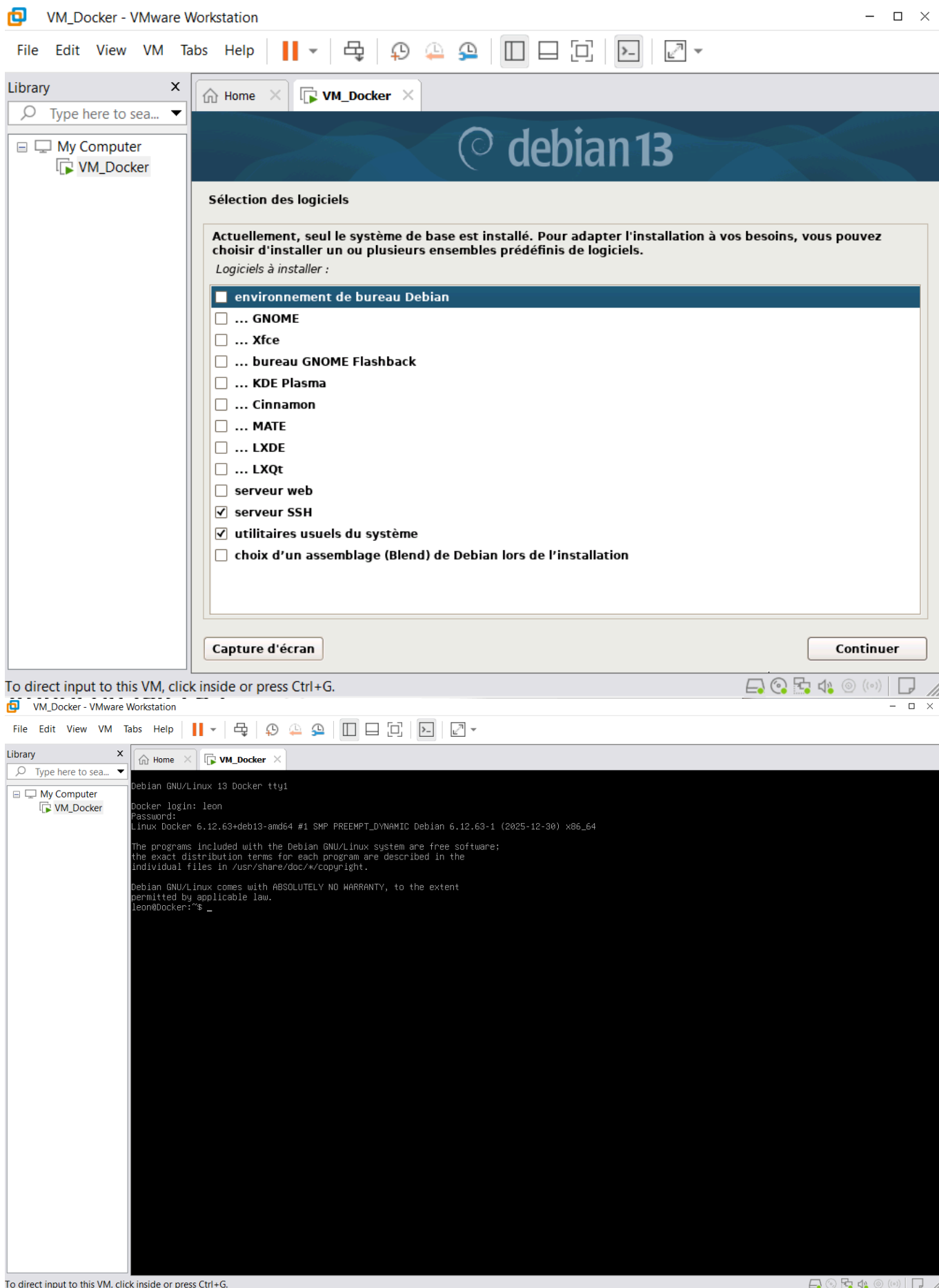


Runtrack_01

JOB01

Etape : Création de la VM



Etape 02 : Installation des dépendances

```
apt install -y ca-certificates curl gnupg
```

Etape 03 : Créer le dossier pour les clés

```
install -m 0755 -d /etc/apt/keyrings
```

Etape 04 : Télécharger la clé GPG Docker

```
curl -fsSL https://download.docker.com/linux/debian/gpg |  
gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

Etape 05 : Créer le dépôt docker

```
echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.gpg]  
https://download.docker.com/linux/debian trixie stable" > /etc/apt/sources.list.d/docker.list
```

Etape 06 : Installer docker

```
apt install -y docker-ce docker-ce-cli containerd.io
```

JOB02 :

Etape 01 : Lancer Docker

```

root@Docker:~# docker --version
Docker version 29.2.1, build a5c7197
root@Docker:~# docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

root@Docker:~# _

```

Etape 02 : Connaître les commandes de bases

- "docker images" pour voir les images
- "docker ps" pour voir les conteneurs actifs
- "docker ps -a" pour voir les conteneurs même arrêtés
- "docker rm ID" pour supprimer les conteneurs
- "docker rmi NOM_IMAGE" pour supprimer une image

Une image est un template immuable contenant le système et l'application.

Un conteneur est un processus isolé basé sur cette image, exécuté grâce au moteur Docker.

```

root@Docker:~# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	ef54e839ef54	25.9kB	9.52kB	U

```

root@Docker:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
root@Docker:~# docker ps -a						
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	
c99b91f07642	hello-world	"/hello"	11 minutes ago	Exited (0)	11 minutes ago	

```

root@Docker:~#

```

JOB03 :

mkdir job03

cd job03

nano Dockerfile

FROM debian:stable-slim

CMD ["echo", "Hello World depuis mon image Debian personnalisée !"]

```
GNU nano 8.4
FROM debian:stable-slim

CMD ["echo", "Hello World depuis mon image Debian personnalisée !"]_
```

docker build -t my-hello .

```
root@Docker:~/job03# docker build -t my-hello .
[+] Building 4.2s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 131B
=> [internal] load metadata for docker.io/library/debian:stable-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/1] FROM docker.io/library/debian:stable-slim@sha256:4448d44b91bf4a13cb1b4e02d9d5f87ed40621d6e33f0ae7b6ddf71d57e29364
=> => resolve docker.io/library/debian:stable-slim@sha256:4448d44b91bf4a13cb1b4e02d9d5f87ed40621d6e33f0ae7b6ddf71d57e29364
=> => sha256:9cd6a97ab1ca6826f63cd20e16474ffdbf6c3bb723669789edf1be9e455afb9d 29.78MB / 29.78MB
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:899aadb4499ab78c8a9103ca0de306aeb76fc3f9b9de6b838a5478492b51db0f
=> => exporting config sha256:9f12bada6bb4a0ef9c0ff42a73faacc71a43e92d81ce760c7e09d9c371b779f6
=> => exporting attestation manifest sha256:cb0e13f7dc263d6bee617f445d6c7743e0a6af106ae2fb14528bc345ff5bbfad
=> => exporting manifest list sha256:ca20521494ec14223049d5fdc66b095f67a2538cbbd4aa77538fd5c4a9952a21
=> => naming to docker.io/library/my-hello:latest
=> => unpacking to docker.io/library/my-hello:latest
root@Docker:~/job03#
```

JOB04

nano dockerfile

```
FROM debian:stable-slim

# Installer SSH
RUN apt update && apt install -y openssh-server && mkdir -p /var/run/ssh

# Définir mot de passe root
RUN echo 'root:root123' | chpasswd

# Autoriser login root
RUN sed -i 's/#PermitRootLogin.*/PermitRootLogin yes/' /etc/ssh/sshd_config && \
    sed -i 's/#PasswordAuthentication.*/PasswordAuthentication yes/' /etc/ssh/sshd_config && \
    sed -i 's/#UsePAM yes/UsePAM no/' /etc/ssh/sshd_config

# Exposer port 22
EXPOSE 22

# Lancer SSH en foreground
CMD ["/usr/sbin/sshd", "-D"]
```

docker build -t my-ssh .

docker images

```
docker run -d -p 2222:22 --name ssh-container my-ssh
```

```
ssh root@localhost -p 2222
```

password: root123

```
root@Docker:~/job04# docker run -d -p 2222:22 --name ssh-container my-ssh
1114cf31c2e67a402cc4051d4148b376f02e90001b95e87535933a8ad18aa0f7
root@Docker:~/job04# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	ef54e839ef54	25.9kB	9.52kB	U
my-hello:latest	ca20521494ec	117MB	29.8MB	U
my-ssh:latest	97f5a16a6d3a	219MB	60.4MB	U

```
root@Docker:~/job04# ssh root@localhost -p 2222
The authenticity of host '[localhost]:2222 ([::1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:SDmspA4xa9WMKxb5HZ8Vvk0HaJZvwZG4btfe30nbDa78.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.
root@localhost's password:
root@1114cf31c2e6:~#
```

```
root@1114cf31c2e6:~# ps aux | grep sshd
root      1  0.0  0.4 11764 8072 ?        Ss   09:48   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root     16  0.0  0.5 17192 11248 ?        Ss   10:09   0:00 sshd-session: root [priv]
root     18  0.1  0.3 17712  7800 ?        S    10:09   0:00 sshd-session: root@pts/0
root     27  0.0  0.0   3504 1828 pts/0    S+   10:12   0:00 grep sshd
root@1114cf31c2e6:~# _
```

JOB05

```

GNU nano 8.4 /root
# ~/.bashrc: executed by bash(1) for non-login shells.

# Note: PS1 is set in /etc/profile, and the default umask is defined
# in /etc/login.defs. You should not need this unless you want different
# defaults for root.
# PS1='${debian_chroot:+($debian_chroot)}\h:\w\$ '
# umask 022

# You may uncomment the following lines if you want `ls' to be colorized:
# export LS_OPTIONS='--color=auto'
# eval "$(dircolors)"
# alias ls='ls $LS_OPTIONS'
# alias ll='ls $LS_OPTIONS -l'
# alias l='ls $LS_OPTIONS -lA'
#
# Some more alias to avoid making mistakes:
# alias rm='rm -i'
# alias cp='cp -i'
# alias mv='mv -i'

alias dps='docker ps'
alias dpsa='docker ps -a'
alias di='docker images'
alias drm='docker rm -f'
alias dri='docker rmi'
alias dr start='docker start'
alias dstop='docker stop'
alias dlogs='docker logs -f'
alias dexec='docker exec -it'
alias drun='docker run -it --rm'

```

Alias	Commande réelle	Sert à quoi ?	Quand l'utiliser ?
dps	docker ps	Voir les conteneurs en cours d'exécution	Vérifier ce qui tourne
dpsa	docker ps -a	Voir tous les conteneurs (même arrêtés)	Debug / nettoyage
di	docker images	Voir les images stockées localement	Vérifier les images disponibles
drm	docker rm -f	Supprimer un conteneur (force si actif)	Nettoyer un conteneur
dri	docker rmi	Supprimer une image	Nettoyer les images inutiles
dstart	docker start	Redémarrer un conteneur arrêté	Relancer un service
dstop	docker stop	Arrêter un conteneur proprement	Stopper un service
dlogs	docker logs -f	Voir les logs en temps réel	Debug d'un conteneur

Alias	Commande réelle	Sert à quoi ?	Quand l'utiliser ?
dexec	<code>docker exec -it</code>	Entrer dans un conteneur actif	Administration interne
drun	<code>docker run -it -rm</code>	Lancer un conteneur temporaire interactif	Test rapide

```
root@Docker:~# dps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
1114cf31c2e6	my-ssh	"/usr/sbin/sshd -D"	59 minutes ago	Up 59 minutes	0.0.0.0:2222->22/tcp, [::]:2222->22/tcp	ssh-container

```
root@Docker:~# di
```


IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	ef54e839ef54	25.9kB	9.52kB	U
my-hello:latest	ca20521494ec	117MB	29.8MB	U
my-ssh:latest	97f5a16a6d3a	219MB	60.4MB	U

```
root@Docker:~#
```

Info → U In Use

JOB06

```
root@Docker:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	ef54e839ef54	25.9kB	9.52kB	U
my-hello:latest	ca20521494ec	117MB	29.8MB	U
my-ssh:latest	97f5a16a6d3a	219MB	60.4MB	U

```
root@Docker:~# docker volume create partage
partage
root@Docker:~# ls
job03  job04
root@Docker:~# volume ls
-bash: volume : commande introuvable
root@Docker:~# docker volume ls
DRIVER      VOLUME NAME
local      partage
root@Docker:~#
```



```

root@Docker:~# docker images
IMAGE                ID                DISK USAGE    CONTENT SIZE    EXTRA
hello-world:latest   ef54e839ef54      25.9kB        9.52kB          U
my-hello:latest       ca20521494ec      117MB         29.8MB          U
my-ssh:latest         97f5a16a6d3a      219MB         60.4MB          U
root@Docker:~# docker volume create partage
partage
root@Docker:~# ls
job03  job04
root@Docker:~# volume ls
-bash: volume : commande introuvable
root@Docker:~# docker volume ls
DRIVER      VOLUME NAME
local       partage
root@Docker:~# docker run -it --name c1 -v partage:/data debian bash
Unable to find image 'debian:latest' locally
latest: Pulling from library/debian
866771c43bf5: Pull complete
e6a3c865571e: Download complete
Digest: sha256:3615a749858a1cba49b408fb49c37093db813321355a9ab7c1f9f4836341e9db
Status: Downloaded newer image for debian:latest
root@1851af1124d9:/# echo "Test volume OK" > /data/test.txt
root@1851af1124d9:/# exit
exit
root@Docker:~# docker run -it --name c2 -v partage:/data debian bash
root@b141cf2064b8:/# cat /data/test.txt
Test volume OK
root@b141cf2064b8:/# docker volume ls
bash: docker: command not found
root@b141cf2064b8:/# exit
exit
root@Docker:~# docker volume ls
DRIVER      VOLUME NAME
local       partage
root@Docker:~# docker rm -f c1 c2
c1
c2
root@Docker:~# ls
job03  job04
root@Docker:~# docker volume ls
DRIVER      VOLUME NAME
local       partage
root@Docker:~#

```

Docker sépare l'exécution des applications (conteneurs) du stockage des données (volumes). Plusieurs conteneurs peuvent monter le même volume afin de partager des données. Les volumes sont persistants : ils ne sont pas supprimés lorsque les conteneurs sont supprimés.

JOB07

```

root@Docker:~/job07# ls
docker-compose.yml
root@Docker:~/job07# cat docker-compose.yml
services:
  nginx:
    image: nginx:alpine
    container_name: nginx
    ports:
      - "8080:80"
    volumes:
      - webdata:/usr/share/nginx/html
    depends_on:
      - ftp
  ftp:
    image: fauria/vsftpd
    container_name: ftp
    ports:
      - "2121:21"
      - "30000-30009:30000-30009"
    environment:
      - FTP_USER=leon
      - FTP_PASS=leon123
      - PASV_ADDRESS=192.168.163.129
      - PASV_MIN_PORT=30000
      - PASV_MAX_PORT=30009
    volumes:
      - webdata:/usr/share/nginx/html

volumes:
  webdata:
root@Docker:~/job07# _

```

```

root@Docker:~/job07# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
debian:latest	3615a749858a	186MB	52.5MB	
fauria/vsftpd:latest	6d71d7c7f1b0	574MB	143MB	U
hello-world:latest	ef54e839ef54	25.9kB	9.52kB	U
my-hello:latest	ca20521494ec	117MB	29.8MB	U
my-ssh:latest	97f5a16a6d3a	219MB	60.4MB	U
nginx:alpine	1d13701a5f9f	93.4MB	26.9MB	U

```

root@Docker:~/job07# _

```

```
root@Docker:~/job07# docker compose up
[+] up 27/27
```

```
♦ Image nginx:alpine      Pulled
♦ Image fauria/vsftpd     Pulled
♦ Network job07_default   Created
♦ Volume job07_webdata    Created
♦ Container ftp           Created
♦ Container nginx         Created
```

Attaching to ftp, nginx

```
ftp | *****
ftp | *
ftp | *      Docker image: fauria/vsftpd      *
ftp | *      https://github.com/fauria/docker-vsftpd  *
ftp | *
ftp | *****
```

SERVER SETTINGS

```
ftp | -----
ftp | • FTP User: leon
ftp | • FTP Password: leon123
ftp | • Log file: /var/log/vsftpd/vsftpd.log
ftp | • Redirect vsftpd log to STDOUT: No.
```

```
nginx | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform c
nginx | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
nginx | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.s
nginx | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/def
nginx | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/d
nginx | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
nginx | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
nginx | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
nginx | /docker-entrypoint.sh: Configuration complete; ready for start up
nginx | 2026/02/25 13:54:32 [notice] 1#1: using the "epoll" event method
nginx | 2026/02/25 13:54:32 [notice] 1#1: nginx/1.29.5
nginx | 2026/02/25 13:54:32 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
nginx | 2026/02/25 13:54:32 [notice] 1#1: OS: Linux 6.12.63+deb13-amd64
nginx | 2026/02/25 13:54:32 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
nginx | 2026/02/25 13:54:32 [notice] 1#1: start worker processes
nginx | 2026/02/25 13:54:32 [notice] 1#1: start worker process 31
nginx | 2026/02/25 13:54:32 [notice] 1#1: start worker process 32
```

☒ Enable Watch ☐ Detach

```
root@Docker:~/job07# docker compose up -d
root@Docker:~/job07# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS
a331ce740c68   nginx:alpine  "/docker-entrypoint..."  6 minutes ago  Up 4 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
d4c57f253dcb   fauria/vsftpd  "/usr/sbin/run-vsftpd..."  6 minutes ago  Up 4 seconds  20/tcp, 0.0.0.0:2121->21/tcp, [::]:2121->21/tcp
1114cf31c2e6   my-ssh      "/usr/sbin/sshd -D"        4 hours ago  Up 4 hours    0.0.0.0:2222->22/tcp, [::]:2222->22/tcp
root@Docker:~/job07#
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Leon Camuzat